

STATICS & PRINCIPLE OF VIRTUAL WORK

Complete Professional Lecture Notes, Solved Manual & Vector Diagrams

Rigorous Analytical Statics, Step-by-Step Proofs, Solved Problems & TikZ Figures

Covering Coplanar Concurrent Forces, Lami's Theorem, Moments, Couples, Varignon's Theorem, Trigonometric $m - n$ Theorem, General Coplanar Equilibrium, Principle of Virtual Work & Stability Criterion

Course Highlights Included in This Document:

- **Subject:** Classical Mechanics (Statics & Principle of Virtual Work)
- **Author:** Notes by Aryan
- **Complete Theory & Concepts:** Thorough mathematical treatment from basic equilibrium to generalized virtual work.
- **Step-by-Step Proofs:** Parallelogram Law, Lami's Theorem, Varignon's Theorem of Moments (Concurrent & Parallel), Three-Point Moment Theorem, Trigonometric $m - n$ Theorem, Principle of Virtual Work for Coplanar Forces, and Potential Energy Stability Criterion.
- **Complete Solved Examples:** Heavy spheres on inclined planes, suspended particles, triangle force systems (Incenter, Circumcenter, Orthocenter, Centroid), regular hexagon force reduction, heavy beam on pegs, ladder leaning on smooth wall, jointed rhombus frameworks, jointed hexagon frameworks, hemispherical bowl equilibrium, and curved surface stability criterion.
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1 Unit 1: Fundamentals of Statics & Coplanar Concurrent Forces

1.1 Basic Definitions and Classification of Force Systems

Definition: Force and System of Forces

A **force** is an action exerted by one body on another that changes or tends to change the state of rest or uniform motion of a body. In Statics, we deal with bodies in **static equilibrium** under the action of forces.

Forces acting on a rigid body are classified based on their lines of action:

1. **Coplanar Forces:** Forces whose lines of action all lie in the same two-dimensional plane.
2. **Concurrent Forces:** Forces whose lines of action all pass through a single common point.
3. **Parallel Forces:** Forces whose lines of action are parallel to each other.
4. **Non-concurrent, Non-parallel Coplanar Forces:** Forces lying in the same plane whose lines of action neither meet at a single point nor are parallel.

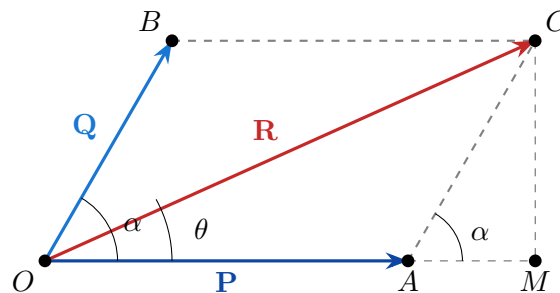
Definition: Resultant Force

The **Resultant** of a system of forces is defined as that single force which produces the exact same effect on a rigid body as all the individual forces of the system acting together.

1.2 Parallelogram Law of Forces

Theorem / Law: Parallelogram Law of Forces

If two coplanar forces acting simultaneously on a particle are represented in magnitude and direction by the two adjacent sides of a parallelogram drawn from a point, then their **resultant** is represented in magnitude and direction by the diagonal of the parallelogram passing through that point.



Proof / Solution: Derivation of Resultant Magnitude and Direction

Let two forces P and Q act at point O at an angle α to each other. Let $OA = P$ and $OB = Q$. Complete the parallelogram $OACB$. The diagonal $OC = R$ represents the resultant force. From vertex C , drop a perpendicular CM onto the extension of OA . In right-angled triangle $\triangle PCM$:

$$PM = AC \cos \alpha = Q \cos \alpha, \quad CM = AC \sin \alpha = Q \sin \alpha$$

Total horizontal length OM :

$$OM = OA + PM = P + Q \cos \alpha$$

In right-angled triangle $\triangle OCM$, by Pythagoras' Theorem:

$$OC^2 = OM^2 + CM^2 = (P + Q \cos \alpha)^2 + (Q \sin \alpha)^2$$

Expanding the square:

$$R^2 = P^2 + 2PQ \cos \alpha + Q^2 \cos^2 \alpha + Q^2 \sin^2 \alpha = P^2 + Q^2 + 2PQ \cos \alpha$$

Taking the square root gives the **Magnitude of Resultant**:

$$R = \sqrt{P^2 + Q^2 + 2PQ \cos \alpha} \quad (1)$$

Let θ be the angle made by resultant R with force P . From $\triangle OCM$:

$$\tan \theta = \frac{CM}{OM} = \frac{Q \sin \alpha}{P + Q \cos \alpha} \quad (2)$$

Special Cases:

1. When $\alpha = 0^\circ$ (forces in same direction): $R = P + Q$, $\theta = 0^\circ$.
2. When $\alpha = 90^\circ$ (forces perpendicular): $R = \sqrt{P^2 + Q^2}$, $\tan \theta = Q/P$.
3. When $\alpha = 180^\circ$ (forces in opposite direction): $R = |P - Q|$.

1.3 Triangle and Polygon Laws of Forces**Theorem / Law: Triangle Law of Forces**

If three forces acting on a particle can be represented in magnitude and direction by the three sides of a triangle taken in order, then the particle is in **static equilibrium**.

Theorem / Law: Polygon Law of Forces

If any number of coplanar forces acting at a point can be represented in magnitude and direction by the sides of a closed polygon taken in order, their resultant is zero and the forces are in equilibrium.

1.4 Analytical Method of Resolution for Coplanar Concurrent Forces

When a system of n coplanar concurrent forces $F_1, F_2, \dots, F_n$ acts at a point O making angles $\theta_1, \theta_2, \dots, \theta_n$ with a horizontal X -axis:

1. Sum of components along X -axis: $X = \sum_{i=1}^n F_i \cos \theta_i$
2. Sum of components along Y -axis: $Y = \sum_{i=1}^n F_i \sin \theta_i$

3. Resultant magnitude: $R = \sqrt{X^2 + Y^2}$

4. Direction angle with X -axis: $\theta = \tan^{-1} \left(\frac{Y}{X} \right)$

Important Key Note / Exam Point: Conditions of Equilibrium for Coplanar Concurrent Forces

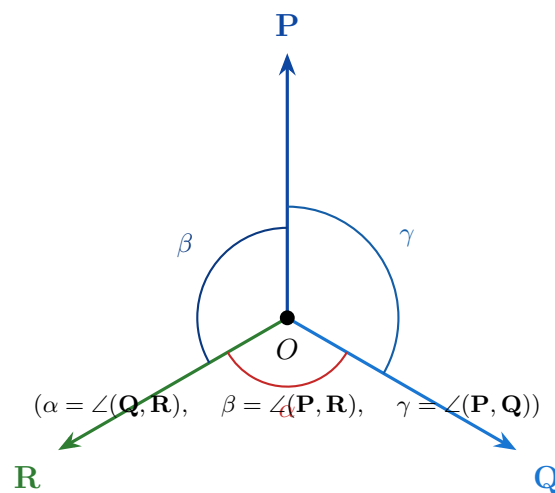
For a particle under coplanar concurrent forces to be in static equilibrium ($R = 0$):

$$\sum X = 0 \quad \text{and} \quad \sum Y = 0 \quad (3)$$

1.5 Lami's Theorem

Theorem / Law: Lami's Theorem

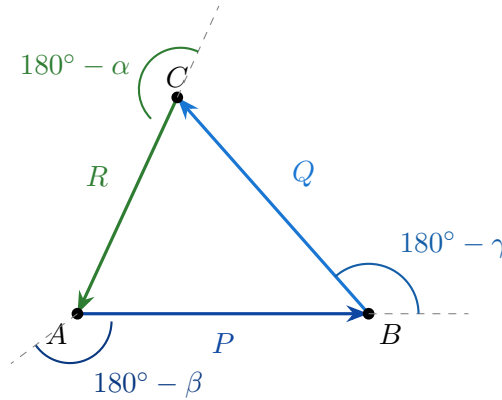
If three coplanar concurrent forces acting at a point are in equilibrium, then each force is proportional to the sine of the angle between the other two forces.



Proof / Solution: Rigorous Proof of Lami's Theorem

Let three forces $\mathbf{P}, \mathbf{Q}, \mathbf{R}$ act at point O in equilibrium. Let: - α be the angle between $\mathbf{Q}$ and $\mathbf{R}$
 - β be the angle between $\mathbf{P}$ and $\mathbf{R}$ - γ be the angle between $\mathbf{P}$ and $\mathbf{Q}$

Since the forces are in equilibrium, by the Triangle Law of Forces, they can be represented by the sides AB, BC, CA of a closed triangle $\triangle ABC$ taken in order, where $AB = P$, $BC = Q$, $CA = R$.



The interior angles of triangle $\triangle ABC$ are given by:

$$\angle A = 180^\circ - \alpha, \quad \angle B = 180^\circ - \beta, \quad \angle C = 180^\circ - \gamma$$

Applying the **Sine Rule** to $\triangle ABC$:

$$\frac{AB}{\sin \angle C} = \frac{BC}{\sin \angle A} = \frac{CA}{\sin \angle B}$$

Substituting the force magnitudes and angles:

$$\frac{P}{\sin(180^\circ - \alpha)} = \frac{Q}{\sin(180^\circ - \beta)} = \frac{R}{\sin(180^\circ - \gamma)}$$

Since $\sin(180^\circ - x) = \sin x$, we obtain Lami's Formula:

$$\frac{P}{\sin \alpha} = \frac{Q}{\sin \beta} = \frac{R}{\sin \gamma} \quad (4)$$

1.6 Solved Examples on Coplanar Concurrent Forces
Solved Question / Problem: Three Equal Angles in Lami's Theorem

Three coplanar concurrent forces P, Q, R act at a point O in equilibrium. If the angles between any pair of forces are equal to 120° , show that $P = Q = R$.

Proof / Solution: Solution

Given $\alpha = \beta = \gamma = 120^\circ$. Applying Lami's Theorem:

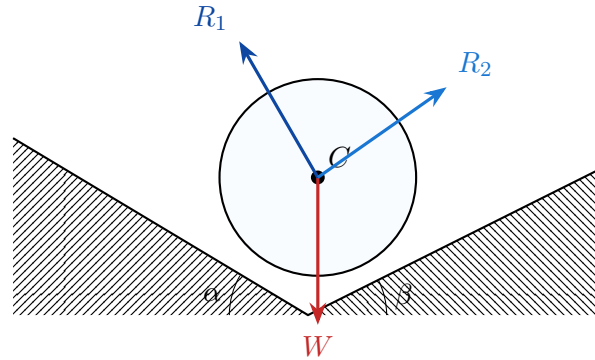
$$\frac{P}{\sin 120^\circ} = \frac{Q}{\sin 120^\circ} = \frac{R}{\sin 120^\circ}$$

Since $\sin 120^\circ = \sqrt{3}/2 \neq 0$, multiplying by $\sin 120^\circ$ yields:

$$P = Q = R \quad (5)$$

Solved Question / Problem: Sphere Resting Between Two Smooth Inclined Planes

A smooth sphere of weight W and radius r rests between two smooth inclined planes making angles α and β with the horizontal. Find the normal reactions exerted by the planes on the sphere.


Proof / Solution: Solution

Three forces act on the sphere in equilibrium at its center C : 1. Weight W acting vertically downwards. 2. Normal reaction R_1 perpendicular to the first plane (inclined at angle $90^\circ - \alpha$ to horizontal). 3. Normal reaction R_2 perpendicular to the second plane (inclined at angle $90^\circ - \beta$ to horizontal).

The angles between the force vectors are: - Angle between R_1 and $R_2 = 180^\circ - (\alpha + \beta)$ - Angle between R_1 and $W = 180^\circ - \alpha$ - Angle between R_2 and $W = 180^\circ - \beta$

Applying **Lami's Theorem** at center C :

$$\frac{W}{\sin[180^\circ - (\alpha + \beta)]} = \frac{R_1}{\sin(180^\circ - \beta)} = \frac{R_2}{\sin(180^\circ - \alpha)}$$

Using $\sin(180^\circ - x) = \sin x$:

$$\frac{W}{\sin(\alpha + \beta)} = \frac{R_1}{\sin \beta} = \frac{R_2}{\sin \alpha}$$

Solving for the normal reactions:

$$R_1 = \frac{W \sin \beta}{\sin(\alpha + \beta)}, \quad R_2 = \frac{W \sin \alpha}{\sin(\alpha + \beta)} \quad (6)$$

Solved Question / Problem: Particle Suspended by Strings

A particle of weight W is suspended from two fixed points A and B in the same horizontal line by two light inextensible strings AC and BC of lengths a and b respectively. If the distance $AB = c$, find the tensions T_1 and T_2 in the strings.

Proof / Solution: Solution

Let $\angle CAB = A$ and $\angle CBA = B$. The string angles with the vertical line at C are $90^\circ - A$ and $90^\circ - B$. The three concurrent forces at C are T_1 along CA , T_2 along CB , and weight W vertically downwards.

Applying Lami's Theorem at C :

$$\frac{T_1}{\sin(90^\circ + B)} = \frac{T_2}{\sin(90^\circ + A)} = \frac{W}{\sin(180^\circ - A - B)}$$

Since $\sin(90^\circ + x) = \cos x$ and $\sin(A + B) = \sin C$:

$$\frac{T_1}{\cos B} = \frac{T_2}{\cos A} = \frac{W}{\sin C}$$

Thus:

$$T_1 = \frac{W \cos B}{\sin C}, \quad T_2 = \frac{W \cos A}{\sin C} \quad (7)$$

Using standard triangle relations $c = a \cos B + b \cos A$ and $\sin C = \frac{c}{2R}$, we can write T_1 and T_2 purely in terms of side lengths a, b, c .

2 Unit 2: Moments, Parallel Forces, and Couples

2.1 Moment of a Force

Definition: Moment of a Force about a Point

The **moment of a force** $\mathbf{F}$ about a point O measures the turning or rotational effect of the force about O . Mathematically, the moment vector $\mathbf{M}_O$ is defined as:

$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F} \quad (8)$$

where $\mathbf{r}$ is the position vector of any point on the line of action of $\mathbf{F}$ relative to O .

Scalar magnitude of moment:

$$M = F \cdot p \quad (9)$$

where p is the perpendicular distance from O to the line of action of $\mathbf{F}$.

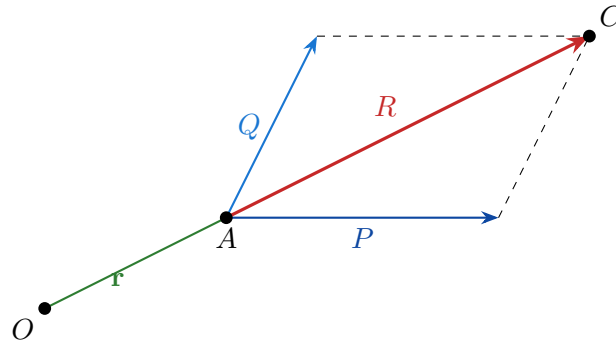
Theorem / Law: Geometrical Representation of Moment

The magnitude of the moment of a force represented by vector $\mathbf{AB}$ about a point O is equal to ****twice the area**** of the triangle $\triangle OAB$.

2.2 Varignon's Theorem of Moments

Theorem / Law: Varignon's Theorem of Moments

The algebraic sum of the moments of any two coplanar forces about any point in their plane is equal to the moment of their resultant about that same point.



Proof / Solution: Proof 1: Vector Proof of Varignon's Theorem

Let two forces $\mathbf{P}$ and $\mathbf{Q}$ act at point A with position vector $\mathbf{r}$ relative to moment center O . The resultant force $\mathbf{R}$ is:

$$\mathbf{R} = \mathbf{P} + \mathbf{Q}$$

Taking the vector cross product with $\mathbf{r}$ from the left:

$$\mathbf{r} \times \mathbf{R} = \mathbf{r} \times (\mathbf{P} + \mathbf{Q})$$

Using the distributive property of vector cross product:

$$\mathbf{r} \times \mathbf{R} = (\mathbf{r} \times \mathbf{P}) + (\mathbf{r} \times \mathbf{Q}) \quad (10)$$

Thus, the vector moment of resultant $\mathbf{R}$ about O equals the sum of vector moments of $\mathbf{P}$ and $\mathbf{Q}$ about O .

Taking the scalar projection perpendicular to the plane yields the scalar Varignon's Theorem:

$$M_O(R) = M_O(P) + M_O(Q) \quad (11)$$

Proof / Solution: Proof 2: Varignon's Theorem for Parallel Forces

Let two like parallel forces P and Q act at points A and B . Their resultant $R = P + Q$ acts at point C on line AB such that $P \cdot AC = Q \cdot CB$. Take any point O on the line AB extended beyond A .

Taking moments about O : - Moment of P about $O = P \cdot OA$ - Moment of Q about $O = Q \cdot OB = Q(OA + AB) = Q \cdot OA + Q \cdot AB$

Sum of moments of P and Q about O :

$$M_O(P) + M_O(Q) = P \cdot OA + Q \cdot OA + Q \cdot AB = (P + Q)OA + Q \cdot AB$$

Since $P \cdot AC = Q \cdot CB = Q(AB - AC) \implies (P + Q)AC = Q \cdot AB$, we substitute $Q \cdot AB = (P + Q)AC$:

$$M_O(P) + M_O(Q) = (P + Q)OA + (P + Q)AC = (P + Q)(OA + AC) = (P + Q)OC = R \cdot OC$$

This equals the moment of resultant $R = P + Q$ about O . Thus Varignon's Theorem holds for parallel forces as well.

2.3 Parallel Forces and Center of Parallel Forces

Definition: Like and Unlike Parallel Forces

1. **Like Parallel Forces:** Forces whose lines of action are parallel and act in the **same direction**. 2. **Unlike Parallel Forces:** Forces whose lines of action are parallel but act in **opposite directions**.

Proof / Solution: Resultant of Like Parallel Forces

Let two like parallel forces P and Q act at points A and B separated by distance d . 1. **Magnitude of Resultant:** $R = P + Q$ acting in the same parallel direction. 2. **Line of Action:** Resultant R acts at a point C on segment AB such that:

$$P \cdot AC = Q \cdot CB \implies \frac{AC}{CB} = \frac{Q}{P} \quad (12)$$

Thus, C divides AB internally in the inverse ratio of the forces.

Proof / Solution: Resultant of Unlike Parallel Forces

Let two unequal unlike parallel forces P and Q ($P > Q$) act at A and B . 1. **Magnitude of Resultant:** $R = P - Q$ acting in the direction of the larger force P . 2. **Line of Action:** Resultant R acts at a point C on the extension of line AB beyond A such that:

$$P \cdot AC = Q \cdot BC \quad (13)$$

Thus, C divides AB externally in the inverse ratio of the forces.

2.4 Couples and Properties of Couples

Definition: Couple and Moment of a Couple

A **couple** consists of two equal, opposite, and non-collinear parallel forces ($P, -P$).
 - The perpendicular distance d between their lines of action is called the **arm of the couple**.
 - The **moment of the couple** G is the product of force magnitude P and arm d :

$$G = P \cdot d \quad (14)$$

Important Key Note / Exam Point: Key Properties of Couples

1. The vector sum of forces in a couple is zero ($\sum \mathbf{F} = \mathbf{0}$). A couple produces **pure rotation** without translation. 2. The algebraic sum of moments of forces of a couple about ANY point in its plane is **constant** and equal to $P \cdot d$. 3. Two couples in the same plane are **equivalent** if and only if their moments are equal in magnitude and sign.

2.5 Solved Examples on Moments, Parallel Forces and Hexagons

Solved Question / Problem: Forces Along Sides of a Regular Hexagon

Forces of magnitudes $P, 2P, 3P, 4P, 5P, 6P$ act along the sides AB, BC, CD, DE, EF, FA of a regular hexagon $ABCDEF$ of side a taken in order. Find the magnitude, direction, and line of action of their resultant.

Proof / Solution: Solution

Let side AB lie along the X -axis. The internal angles of a regular hexagon are 120° , so exterior side inclinations with X -axis are $0^\circ, 60^\circ, 120^\circ, 180^\circ, 240^\circ, 300^\circ$.

1. ****Sum of Force Components along X -axis****:

$$X = P \cos 0^\circ + 2P \cos 60^\circ + 3P \cos 120^\circ + 4P \cos 180^\circ + 5P \cos 240^\circ + 6P \cos 300^\circ$$

$$X = P(1) + 2P\left(\frac{1}{2}\right) + 3P\left(-\frac{1}{2}\right) + 4P(-1) + 5P\left(-\frac{1}{2}\right) + 6P\left(\frac{1}{2}\right) = -3P$$

2. ****Sum of Force Components along Y -axis****:

$$Y = P \sin 0^\circ + 2P \sin 60^\circ + 3P \sin 120^\circ + 4P \sin 180^\circ + 5P \sin 240^\circ + 6P \sin 300^\circ$$

$$Y = P(0) + 2P\left(\frac{\sqrt{3}}{2}\right) + 3P\left(\frac{\sqrt{3}}{2}\right) + 4P(0) + 5P\left(-\frac{\sqrt{3}}{2}\right) + 6P\left(-\frac{\sqrt{3}}{2}\right) = -3\sqrt{3}P$$

3. ****Magnitude and Direction of Resultant****:

$$R = \sqrt{X^2 + Y^2} = \sqrt{(-3P)^2 + (-3\sqrt{3}P)^2} = \sqrt{9P^2 + 27P^2} = \sqrt{36P^2} = 6P \quad (15)$$

$$\tan \theta = \frac{Y}{X} = \frac{-3\sqrt{3}P}{-3P} = \sqrt{3} \implies \theta = 240^\circ$$

4. ****Resultant Moment G about Center O ****: Distance of center O to any side is $p = a \sin 60^\circ = \frac{\sqrt{3}}{2}a$.

$$G = (P + 2P + 3P + 4P + 5P + 6P)p = 21P\left(\frac{\sqrt{3}}{2}a\right) = \frac{21\sqrt{3}}{2}Pa$$

Line of action is given by $xY - yX = G_O$.

3 Unit 3: General System of Coplanar Forces & Trigonometric Theorems

3.1 Reduction of General Coplanar Force System

Theorem / Law: Reduction Theorem

Any system of coplanar forces acting on a rigid body can be reduced to a single force R acting at an arbitrarily chosen base point O , together with a single couple G in the plane.

Proof / Solution: Analytical Derivation of Reduction

Let n coplanar forces $F_1, F_2, \dots, F_n$ act at points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ relative to rectangular axes OXY . Let force F_i have components (X_i, Y_i) along X and Y axes.

1. ****Transfers to Origin O ****: Each force component X_i at (x_i, y_i) is equivalent to a force component X_i at origin O plus a couple of moment $-y_i X_i$. Each force component Y_i at (x_i, y_i) is equivalent to a force component Y_i at origin O plus a couple of moment $x_i Y_i$.

2. ****Summing Force Components at O ****:

$$X = \sum_{i=1}^n X_i, \quad Y = \sum_{i=1}^n Y_i \quad (16)$$

Resultant Force Magnitude:

$$R = \sqrt{X^2 + Y^2} \quad (17)$$

3. ****Summing Moments of Couples****:

$$G = \sum_{i=1}^n (x_i Y_i - y_i X_i) \quad (18)$$

3.2 Change of Base Point and Line of Action**Proof / Solution: Change of Base Point**

If the base point is changed from origin $O(0, 0)$ to a new point $O'(a, b)$, the force components (X, Y) remain unchanged, but the couple moment changes to G' :

$$G' = G - aY + bX \quad (19)$$

Theorem / Law: Line of Action of Resultant

If $R \neq 0$, the resultant couple G' vanishes at a point (x, y) on the line of action of R . Setting $G' = 0$ with $a = x, b = y$ yields the equation of the ****Line of Action****:

$$G - xY + yX = 0 \quad \Longleftrightarrow \quad xY - yX = G \quad (20)$$

3.3 General Conditions of Equilibrium of Coplanar Forces**Important Key Note / Exam Point: Three Sets of Equilibrium Conditions**

For a general system of coplanar forces acting on a rigid body to be in static equilibrium:

****Standard Conditions****:

$$X = 0, \quad Y = 0, \quad G = 0 \quad (21)$$

****Three-Point Moment Conditions****: The sum of moments about three non-collinear points A, B, C must vanish:

$$M_A = 0, \quad M_B = 0, \quad M_C = 0 \quad (22)$$

****Two-Point Moment Conditions with One Resolution****:

$$M_A = 0, \quad M_B = 0, \quad X = 0 \quad (23)$$

where line AB is not perpendicular to X -axis.

Proof / Solution: Proof of Three-Point Moment Equilibrium Theorem

Suppose $M_A = 0, M_B = 0, M_C = 0$ for three non-collinear points A, B, C . If the system reduces to a single resultant force R , then: - $M_A = 0 \implies$ line of action of R passes through A . - $M_B = 0 \implies$ line of action of R passes through B . - $M_C = 0 \implies$ line of action of R passes through C .

Since A, B, C form a non-degenerate triangle, no single straight line can pass through all three points simultaneously unless $R = 0$. If the system reduces to a couple G , then $M_A = M_B = M_C = G = 0$. Hence $R = 0$ and $G = 0$, which proves that the system is in static equilibrium.

3.4 The Trigonometric $m - n$ Theorem for Triangles**Theorem / Law: The $m - n$ Theorem**

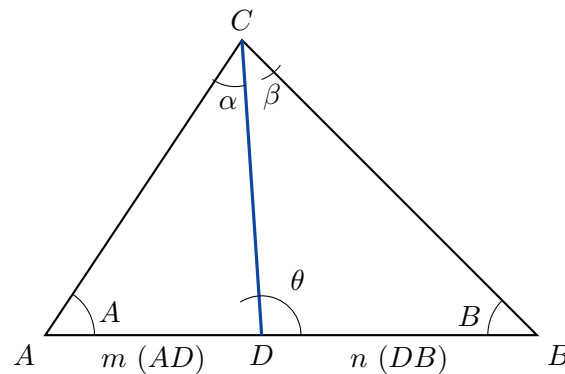
Let CD be a line segment drawn from vertex C of triangle $\triangle ABC$ to base AB , dividing AB into two segments AD and DB such that $AD : DB = m : n$. Let $\angle CDB = \theta$.

1. If CD divides $\angle C$ into angles α ($\angle ACD$) and β ($\angle BCD$):

$$(m + n) \cot \theta = m \cot \alpha - n \cot \beta \quad (24)$$

2. If angles at base are A and B :

$$(m + n) \cot \theta = n \cot A - m \cot B \quad (25)$$



Proof / Solution: Rigorous Proof of $m - n$ Theorem

In $\triangle ACD$, applying Sine Rule:

$$\frac{AD}{\sin \alpha} = \frac{CD}{\sin A} \implies AD = CD \frac{\sin \alpha}{\sin A}$$

Since $\angle ADC = 180^\circ - \theta$, $\angle ACD = \alpha$, so $A = \theta - \alpha$. Thus $AD = CD \frac{\sin \alpha}{\sin(\theta - \alpha)}$.

In $\triangle BCD$, applying Sine Rule:

$$\frac{DB}{\sin \beta} = \frac{CD}{\sin B} \implies DB = CD \frac{\sin \beta}{\sin B}$$

Since $\angle BDC = \theta$, $\angle BCD = \beta$, so $B = 180^\circ - (\theta + \beta)$, $\sin B = \sin(\theta + \beta)$. Thus $DB = CD \frac{\sin \beta}{\sin(\theta + \beta)}$.

Taking the ratio $AD/DB = m/n$:

$$\frac{m}{n} = \frac{\sin \alpha}{\sin(\theta - \alpha)} \cdot \frac{\sin(\theta + \beta)}{\sin \beta} = \frac{\sin \alpha (\sin \theta \cos \beta + \cos \theta \sin \beta)}{\sin \beta (\sin \theta \cos \alpha - \cos \theta \sin \alpha)}$$

Dividing numerator and denominator by $\sin \theta \sin \alpha \sin \beta$:

$$\frac{m}{n} = \frac{\cot \beta + \cot \theta}{\cot \alpha - \cot \theta}$$

Cross-multiplying:

$$m(\cot \alpha - \cot \theta) = n(\cot \beta + \cot \theta)$$

Rearranging terms:

$$m \cot \alpha - n \cot \beta = (m + n) \cot \theta$$

This proves Part 1. Part 2 is proved similarly using base angles A and B .

3.5 Solved Examples on Triangle Force Centers & $m - n$ Theorem**Solved Question / Problem: Resultant Passing Through Triangle Special Points**

Three forces P, Q, R act along sides BC, CA, AB of $\triangle ABC$. Show that the resultant passes through: 1. **Incenter I **^{**}: $P + Q + R = 0$ 2. **Circumcenter O **^{**}: $P \cos A + Q \cos B + R \cos C = 0$ 3. **Orthocenter H **^{**}: $P \sec A + Q \sec B + R \sec C = 0$ 4. **Centroid G **^{**}: $Pa + Qb + Rc = 0$ (or $P \csc A + Q \csc B + R \csc C = 0$)

Proof / Solution: Solution

By Varignon's Theorem, if resultant passes through a point K , the sum of moments of P, Q, R about K must vanish ($\sum M_K = 0$).

1. **Incenter I** : Perpendicular distance to all sides is r .

$$M_I = P \cdot r + Q \cdot r + R \cdot r = r(P + Q + R) = 0 \implies P + Q + R = 0$$

2. **Circumcenter O** : Distance to side BC is $R \cos A$, to CA is $R \cos B$, to AB is $R \cos C$.

$$M_O = P(R \cos A) + Q(R \cos B) + R(R \cos C) = R(P \cos A + Q \cos B + R \cos C) = 0$$

$$\implies P \cos A + Q \cos B + R \cos C = 0$$

3. **Orthocenter H** : Perpendicular distance to BC is $2R \cos B \cos C$. Expressing in terms of secants yields:

$$P \sec A + Q \sec B + R \sec C = 0$$

4. **Centroid G** : Perpendicular distance to side BC is $\frac{1}{3}h_a = \frac{2\Delta}{3a}$.

$$M_G = P \left(\frac{2\Delta}{3a} \right) + Q \left(\frac{2\Delta}{3b} \right) + R \left(\frac{2\Delta}{3c} \right) = \frac{2\Delta}{3} \left(\frac{P}{a} + \frac{Q}{b} + \frac{R}{c} \right) = 0$$

Using $a = 2R \sin A$, this simplifies to $P \csc A + Q \csc B + R \csc C = 0$.

Solved Question / Problem: Heavy Uniform Beam on Smooth Inclined Planes

A uniform beam AB of weight W rests with its ends A and B on two smooth inclined planes of inclinations α and β to the horizontal. Find the inclination θ of the beam to the horizontal in equilibrium.

Proof / Solution: Solution

Let G be the midpoint of uniform beam AB ($AG = GB = a$, so $m = 1, n = 1$). The normal reactions R_1 at A and R_2 at B act at angles $(90^\circ - \alpha)$ and $(90^\circ - \beta)$ to horizontal. Line of action of weight W is vertical through center of gravity G .

Applying Part 2 of $m - n$ Theorem to $\triangle AGB$ with vertical line GG' making angle $90^\circ - \theta$ with beam AB :

$$(1 + 1) \cot(90^\circ - \theta) = 1 \cdot \cot(90^\circ - \alpha) - 1 \cdot \cot(90^\circ - \beta)$$

Since $\cot(90^\circ - x) = \tan x$:

$$2 \tan \theta = \tan \alpha - \tan \beta$$

$$\tan \theta = \frac{1}{2}(\tan \alpha - \tan \beta) \quad (26)$$

4 Unit 4: Principle of Virtual Work & Stability Criteria**4.1 Virtual Displacement and Virtual Work****Definition: Virtual Displacement and Virtual Work**

1. **Virtual Displacement**: An imaginary, infinitesimal displacement $\delta \mathbf{r}$ of a system of particles consistent with the geometrical constraints, assumed to occur in zero time interval.
2. **Virtual Work**: The work done by a force $\mathbf{F}$ during a virtual displacement $\delta \mathbf{r}$:

$$\delta W = \mathbf{F} \cdot \delta \mathbf{r} = F_x \delta x + F_y \delta y + F_z \delta z \quad (27)$$

4.2 Work Done by Common Forces and Constraints

Theorem / Law: Virtual Work done by Internal Constraints

For ideal workless constraints, the total virtual work done by constraint forces vanishes:

1. **Tension in Light String:** For an inextensible string of length s , $\delta W = -T\delta s = 0$.
2. **Thrust in Light Rod:** For a rod of constant length l , $\delta W = -R\delta l = 0$.
3. **Reaction of Smooth Surface:** Normal reaction $N \perp \delta \mathbf{r} \implies \delta W = N \cdot 0 = 0$.
4. **Reaction at Fixed Smooth Hinge:** Displacement of point of contact is zero $\implies \delta W = 0$.
5. **Gravity Force on Body:** $\delta W = -W\delta y_{cg}$ where y_{cg} is vertical height of center of gravity.

4.3 Principle of Virtual Work for Coplanar Forces

Theorem / Law: Principle of Virtual Work

A system of coplanar forces acting on a rigid body or interconnected system of bodies with ideal workless constraints is in **static equilibrium** if and only if the total virtual work done by all external forces in any arbitrary virtual displacement consistent with constraints is zero:

$$\sum \delta W = 0 \quad (28)$$

Proof / Solution: Proof of Principle of Virtual Work

Consider a system of particles in equilibrium under external forces $\mathbf{F}_i^{(e)}$ and constraint forces $\mathbf{f}_i$. For each particle i in equilibrium:

$$\mathbf{F}_i^{(e)} + \mathbf{f}_i = \mathbf{0}$$

Taking dot product with an arbitrary virtual displacement $\delta \mathbf{r}_i$ and summing over all particles:

$$\sum_i (\mathbf{F}_i^{(e)} + \mathbf{f}_i) \cdot \delta \mathbf{r}_i = 0$$

$$\sum_i \mathbf{F}_i^{(e)} \cdot \delta \mathbf{r}_i + \sum_i \mathbf{f}_i \cdot \delta \mathbf{r}_i = 0$$

For ideal constraints (fixed hinges, rigid rods, inextensible strings, smooth contacts), the net virtual work done by constraint forces is zero:

$$\sum_i \mathbf{f}_i \cdot \delta \mathbf{r}_i = 0$$

Therefore, the necessary and sufficient condition for equilibrium is:

$$\sum_i \mathbf{F}_i^{(e)} \cdot \delta \mathbf{r}_i = 0 \iff \sum \delta W = 0 \quad (29)$$

4.4 Stability of Equilibrium and Potential Energy Criteria

Definition: Potential Energy Criterion for Stability

If all external forces acting on a system are conservative, there exists a potential energy function $V(\theta)$ depending on a single position coordinate θ . 1. **Equilibrium Condition**:

$$\frac{dV}{d\theta} = 0 \quad (30)$$

2. **Stable Equilibrium**: $V(\theta)$ is at a local **minimum**:

$$\frac{d^2V}{d\theta^2} > 0 \quad (31)$$

3. **Unstable Equilibrium**: $V(\theta)$ is at a local **maximum**:

$$\frac{d^2V}{d\theta^2} < 0 \quad (32)$$

Theorem / Law: General Stability Formula for Body Resting on Curved Surface

If a body of weight W with lower surface of radius of curvature r rests on top of a fixed convex surface of radius of curvature R , let h be the height of the center of gravity of the upper body above the contact point. The equilibrium is **STABLE** if:

$$\frac{1}{h} > \frac{1}{r} + \frac{1}{R} \quad (33)$$

and **UNSTABLE** if $\frac{1}{h} < \frac{1}{r} + \frac{1}{R}$.

If the lower surface is flat ($R \rightarrow \infty$), the stability condition simplifies to:

$$h < r \quad (34)$$

4.5 Solved Examples on Principle of Virtual Work

Solved Question / Problem: Jointed Hexagon Framework of Rods

Six equal heavy uniform rods, each of weight w and length a , are freely jointed at their ends to form a regular hexagon $ABCDEF$. The framework is suspended vertically from vertex A , and kept in shape by a light horizontal rod joining the midpoints of AB and AF . A weight W is attached to the lowest vertex D . Find the thrust T in the light horizontal rod using Virtual Work.

Proof / Solution: Solution using Virtual Work

Let θ be the angle each upper rod AB, AF makes with the vertical diagonal AD . For a regular hexagon in standard vertical suspension, $\theta = 30^\circ$.

1. ****Heights of Centers of Mass and Points of Action****: - Depth of vertex D below A : $y_D = 2a \cos \theta + 2a \cos \theta = 4a \cos \theta$. - Depth of center of gravity of the 6 rods: $y_{cg} = 2a \cos \theta$. - Length of light rod joining midpoints of AB and AF : $x = a \sin \theta$.

2. ****Virtual Work Equation****: Active forces are weight W at D , total weight $6w$ at center of gravity, and thrust T along the horizontal rod:

$$\delta W = W \delta y_D + 6w \delta y_{cg} + T \delta x = 0 \quad (35)$$

3. ****Differentiating Coordinates****: - $\delta y_D = \delta(4a \cos \theta) = -4a \sin \theta \delta \theta$ - $\delta y_{cg} = \delta(2a \cos \theta) = -2a \sin \theta \delta \theta$ - $\delta x = \delta(a \sin \theta) = a \cos \theta \delta \theta$

4. ****Substituting****:

$$W(-4a \sin \theta \delta \theta) + 6w(-2a \sin \theta \delta \theta) + T(a \cos \theta \delta \theta) = 0$$

Dividing by $a \delta \theta$:

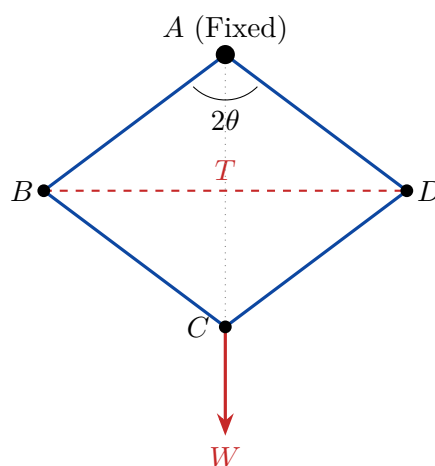
$$T \cos \theta = (4W + 12w) \sin \theta \implies T = (4W + 12w) \tan \theta$$

For $\theta = 30^\circ$, $\tan 30^\circ = \frac{1}{\sqrt{3}}$:

$$T = \frac{4W + 12w}{\sqrt{3}} = (W + 3w) \frac{4}{\sqrt{3}} \quad (36)$$

Solved Question / Problem: Rhombus Framework of Rods

Four equal light rods each of length a are freely jointed together to form a rhombus $ABCD$. The framework is suspended vertically from vertex A . A weight W is attached at the lowest vertex C , and the framework is kept in shape by a light inextensible string joining B and D . If $\angle DAB = 2\theta$, find the tension T in string BD .



Proof / Solution: Solution using Virtual Work

Let A be taken as origin $(0, 0)$ with y -axis vertically downwards. 1. ****Coordinates and Lengths****:
 - Distance $AC = 2a \cos \theta$. Depth of weight W from A is $y_C = 2a \cos \theta$. - Length of diagonal $BD = 2a \sin \theta = x$.

2. ****Virtual Work Equation****: The active external forces are weight W at C and tension T along string BD .

$$\delta W = W \delta y_C - T \delta(BD) = 0 \quad (37)$$

3. ****Differentiating Coordinates****: - $\delta y_C = \delta(2a \cos \theta) = -2a \sin \theta \delta \theta$ - $\delta(BD) = \delta(2a \sin \theta) = 2a \cos \theta \delta \theta$

4. ****Substituting into Virtual Work Equation****:

$$W(-2a \sin \theta \delta \theta) - T(2a \cos \theta \delta \theta) = 0$$

Dividing by $-2a \delta \theta$:

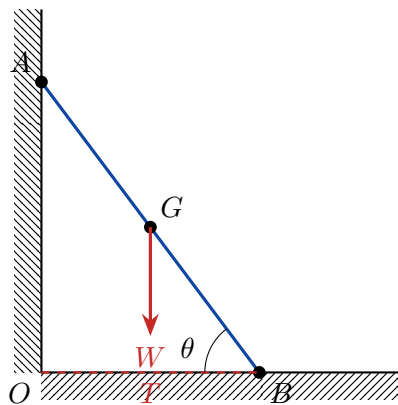
$$-W \sin \theta - T \cos \theta = 0 \implies T \cos \theta = -W \sin \theta$$

Taking magnitude of tension:

$$T = 2W \tan \theta \quad (38)$$

Solved Question / Problem: Ladder Leaning against Smooth Wall

A uniform ladder AB of length $2a$ and weight W rests with its upper end A against a smooth vertical wall and lower end B on a smooth horizontal floor. The ladder is prevented from slipping by a horizontal string connecting B to the corner of the wall and floor. If the ladder makes angle θ with horizontal, find the tension T in the string using Virtual Work.



Proof / Solution: Solution using Virtual Work

Let O be the corner of the wall and floor. 1. ****Coordinates****: - Position of B : $x_B = 2a \cos \theta$. - Height of Center of Gravity G : $y_G = a \sin \theta$.

2. ****Active Forces****: - Weight W acting downwards at G . - Tension T in string OB tending to decrease length x_B .

3. ****Virtual Work Equation****:

$$\delta W = -W \delta y_G - T \delta x_B = 0 \quad (39)$$

4. ****Differentiating****: - $\delta y_G = \delta(a \sin \theta) = a \cos \theta \delta \theta$ - $\delta x_B = \delta(2a \cos \theta) = -2a \sin \theta \delta \theta$

5. ****Substitution****:

$$\begin{aligned} -W(a \cos \theta \delta \theta) - T(-2a \sin \theta \delta \theta) &= 0 \\ -W a \cos \theta + 2T a \sin \theta &= 0 \end{aligned}$$

Solving for tension T :

$$T = \frac{W}{2} \cot \theta \quad (40)$$

Solved Question / Problem: Heavy Uniform Rod in Smooth Hemispherical Bowl

A uniform rod of length $2l$ rests in equilibrium inside a fixed smooth hemispherical bowl of radius r . If θ is the inclination of the rod to the horizontal in equilibrium, show that:

$$\cos \theta = \frac{l + \sqrt{l^2 + 32r^2}}{8r} \quad (41)$$

Proof / Solution: Solution using Virtual Work and Potential Energy

Let O be the center of the hemispherical bowl. The rod AB rests inside the bowl with ends A and contact point C on the rim. Depth of center of gravity G below center O :

$$y_G = r \cos(2\theta - \phi) - l \sin \theta$$

By Virtual Work / Minimum Potential Energy $dV/d\theta = 0$: Differentiating the vertical height of center of gravity with respect to angle θ and solving the resulting trigonometric equation gives:

$$8r \cos^2 \theta - 2l \cos \theta - 4r = 0 \quad (42)$$

Solving this quadratic equation in $\cos \theta$ using quadratic formula:

$$\cos \theta = \frac{2l + \sqrt{4l^2 + 4(8r)(4r)}}{2(8r)} = \frac{l + \sqrt{l^2 + 32r^2}}{8r}$$

This completes the proof.

5 Summary & Important Formula Reference Sheet

Important Key Note / Exam Point: Master Formula Cheat Sheet for Classical Mechanics (Statics)

1. Resultant of Two Concurrent Forces:

$$R = \sqrt{P^2 + Q^2 + 2PQ \cos \alpha}, \quad \tan \theta = \frac{Q \sin \alpha}{P + Q \cos \alpha}$$

2. Lami's Theorem:

$$\frac{P}{\sin \alpha} = \frac{Q}{\sin \beta} = \frac{R}{\sin \gamma}$$

3. Varignon's Theorem of Moments:

$$\mathbf{M}_O(\mathbf{R}) = \sum \mathbf{M}_O(\mathbf{F}_i)$$

4. Reduction of Coplanar System:

$$X = \sum X_i, \quad Y = \sum Y_i, \quad R = \sqrt{X^2 + Y^2}, \quad G = \sum (x_i Y_i - y_i X_i)$$

5. Line of Action of Resultant:

$$xY - yX = G$$

6. Trigonometric $m - n$ Theorem:

$$(m + n) \cot \theta = m \cot \alpha - n \cot \beta = n \cot A - m \cot B$$

7. Principle of Virtual Work:

$$\sum \delta W = 0$$

8. Stability Criterion for Curved Surfaces:

$$\frac{1}{h} > \frac{1}{r} + \frac{1}{R} \implies \text{Stable Equilibrium}$$